

TS-2068 PICO INTERFACE

INTERFACE MANUAL

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**** CONFIDENCIAL ****

(PRELIMINAR DRAFT VERSION 1.1)

INTRODUCTION

This document details the communication protocol specifications between the operating system of the TS-2068 computer and the peripherals that support functionalities of storage and communication.

The new basic commands added to the traditional ones use the protocol TPI ("Timex Protocol Interface"). In this document you will find examples of these extended commands in Basic (called TPI commands from now on) and the detail of the low-level protocol data exchange for each one of them.

For more details on the use of the commands and functionalities, please consult the TS-PICO user manual which explains with examples, the implementation of this protocol in the distribution version.

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The author is not responsible for any damage or harm caused by the use of the information contained at this document. However, if you find any error or consider that a functionality should be added or improved, please contact me at panegustavo@yahoo.com.ar with the subject TPI ERROR or TPI CHANGE REQUEST (these observations will be analyzed and could be included at further releases of this document).

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CHAPTER 1

The new implemented commands are similar to those existing in traditional BASIC, the difference is that functionally they adapt to this new interface.

Example: How to copy a file from tape to SD memory

```
10 SAVE "TPI:SDCARD"           // Select SD port as Default
20 LOAD "TPI:FILE-001.TAP"      // Create "FILE-001.TAP" on SD
30 SAVE "TPI:TAPE"              // Select TAPE port as Default
40 LOAD "123.COD" CODE 40000,100 // Load Code Block From TAPE
50 SAVE "TPI:SDCARD"           // Select SD port as Default
60 SAVE "123.COD" CODE 40000,100 // Save Code Block to SD into .TAP File
70 SAVE "TPI:CLOSE"             // Close .TAP File
80 STOP
```

The TS-PICO system variables are located in the memory area defined as VARS2, from address 05DCDh to address 05DDCh.

This area must be reserved and must not be modified if the TS-PICO system is used.

These variables cannot be relocated if you wish to use the TS-PICO's BASIC functions. A machine code program must be written in RAM, including low-level I/O communication, if this memory area needs to be used.

TPMODE Located in 24027/(5DDB)

```
;-----
; Check Local Commands        VALUE
;-----
; SAVE "TPI:TAPE"            +0 Standart Tape Storage
; SAVE "TPI:PICOPT"          +1 PICO Virtual Printer Channel
; SAVE "TPI:SDCARD"          +2 SD Storage
; SAVE "TPI:TS2040"          +0 Standart Printer Channel
;-----
; SAVE "TPI:FMODE=TAP"        +0 Default Mode
; SAVE "TPI:FMODE=DOS"        +4 MSDOS 3.3 (8.3 FILE NAME) / WITHOUT HEADER
; SAVE "TPI:FMODE=TPI"        +8 WINDOWS (LONG NAMES FILE NAME) / WITH HEADER
; SAVE "TPI:FMODE=FDD"        +12 CP/M - TOS (8.3 FILE NAME) / WITH HEADER
; SAVE "TPI:FMODE=RAW"        +16 WINDOWS (LONG NAMES ) / WITHOUT HEADER
;-----
```

CHAPTER 2

It is possible to use the interface from machine code, the traditional SAVE and LOAD entry points of the TS2068 EXROM memory have been kept for compatibility reasons with existing programs.

The machine code examples are written to use the ZEUS Assembler compiler.

The following assembly language example must be compiled with ZEUS, and the screen record routine will be located at address 40000. In this case it is executed from a BASIC program that must be written and executed with the compiled routine in memory.

EXAMPLE OF A PROGRAM IN MACHINE CODE TO STORE THE SCREEN IN THE SD MEMORY CALLED FROM BASIC.

PROGRAM IN TIMEX BASIC

```
10 CLEAR 39999
20 FOR N=0 TO 22
30 PRINT N,N
40 NEXT N
50 CIRCLE (127,95,20)
60 CIRCLE (127,95,50)
70 SAVE "TPI:SDCARD"           // Select SD port as Default
80 LOAD "TPI:SCREEN-001.TAP"    // Create "SCREEN-001.TAP" on SD
                                // if not exist
70 RANDOMIZE USR 40000         // Calling Save Screen Routine
80 STOP
```

PROGRAM EN ZEUS ASSEMBLER

```
00010  ORG 40000                ; Program Memory Address
00020  ;
00030  DKHSPT EQU  #F4          ; DOCK Horizontal Select Port
00040  HREXPT EQU  #FF          ; Home ROM Extension Select Port
00050  HTYPE  EQU  #03          ; Header Type ID
00060  BTYPE  EQU  #FF          ; Block Type ID
00070  TPMODE EQU  #5DDB        ; TP_MODE (24027)
00080  WTAPE  EQU  #0068        ; EXROM Write Routine
00090  RTAPE  EQU  #00FC        ; EXROM Read Routine
00100  DFILE0 EQU  #4000        ; SCREEN 0 Start Address
00110  DFLEN  EQU  #1B00        ; SCREEN Area Size
00120  ;
00130  ENT                    ; Entry Point (For Zeus X Command)
00140  ;
00150  SASCR: PUSH HL
00160          LD LD IX,HEADER  ; IX=Start Memory Address
00170          LD DE, 17        ; DE= Block LEN
00180          LD A, HTYPE      ; A= CODE TYPE
00190  ;
00200          LD (IX),A        ; Set Header Block type (SCREEN)
00210          LD A,0           ; A=Tipo de Bloque (0=HEADER FF=DATA)
```

```

00220      LD HL, DFILE0
00230      LD (HDADD),HL      ; Set Header Starting Address
00240      LD HL, DFLEN
00250      LD (HDLEN),HL      ; Set Header Block LEN
00260      LD HL, 0
00270      LD (HDVARS),HL      ; Set Header Block HDVARS = 0
00280      LD A,L              ; A=0 Set Header Block
00290      CALL MYSAVE         ; Calling SAVE Rutina
00300 ;
00310      LD IX,DFILE0         ; IX=Screen Address (16384)
00320      LD DE, DFLEN         ; DE= Largo del Bloque (6912)
00330      LD A, BTYPE         ; A=Tipo de Bloque (00=HEADER FF=DATA)
00340 ;
00350      CALL MYSAVE         ; Calling SAVE Rutina
00360 ;
00370      POP HL               ; Restore HL
00375      LD BC, DFLEN        ; Return Bytes SAVED
00380      RET                 ; Return to Caller
00390 ;
00400 MYSAVE: DI              ; Disable Interrupt
00410      PUSH AF             ; Save AF en el STACK
00420      IN A,(HREXPT)       ; A=Valor de HREXPT
00430      LD (PORTFF),A       ; Save Value for Restore Later
00440      SET 7,A              ; Set EXROM Control Bit
00450      OUT (HREXPT),A       ; Enable EXROM
00460      IN A,(DKHSPT)       ; A=Valor de DKHSPT
00470      LD (PORTF4),A       ; Save Value for Restore Later
00480      LD A,3               ; A=3 Indica EXROM en USO
00490      OUT (PORTF4),A       ; Enable EXROM in Chunk 0
00500      LD (TPMODE),A       ; Flag de SD Card en USO
00510      POP AF              ; Restore AF
00520      CALL WTape          ; Calling W_TAPE Routine in ROM
00530      PUSH AF
00540      LD A, (PORTF4)
00550      OUT (PORTF4),A       ; Restore Original HS Value
00560      LD A, (PORTFF)
00570      OUT (HREXPT),A       ; Restore Original HREXPT Value
00580      POP AF
00590      RET
00600 ;
00610 PORTF4: DEFB 0           ; Save TEMP DKHSPT Port Value
00620 PORTFF: DEFB 0           ; Save TEMP HREXPT Port Value
00630 ;
00640 ; STANDARD TAPE HEADER
00650 ;
00660 HEADER:
00670 HDTYPE: DEFB 0
00680 HDNAME: DEFB "MYTEST.COD"
00690 HDLEN: DEFW 0
00700 HDADD: DEFW 0
00710 HDVARS: DEFW 0

```

CHAPTER 3

MACHINE CODE INTRODUCTION

The control routines of the TS-PICO interface are located in the EXROM memory of the computer, for this reason it is necessary to make a memory switch before calling the addresses where the routines are located. To perform this task, you can use the standard call routines of the TS-2068 designed for this purpose, for example, we use GOTO BANK and BANK ENABLE.

The routines available in the TS-PICO BIOS are located in the EXROM.

G_MODE	EQU #1840	; Get Output Mode (Return Mode in BC)
S_MODE	EQU #1842	; Set Output Mode (A = Output Mode)
G_VERS	EQU #1844	; Get TS-PICO Bios Version (Return Version in BC)
G_VERS	EQU #1844	; Get TS-PICO Bios Version (Return Version in BC)
TX_A	EQU #1846	; Transmit Character in A
RX_A	EQU #1848	; Receive Character in A
C_END	EQU #184A	; Command End

; Return Z No Error
; Return NZ Error Code in A
; Return Cy Error I/O

WF_NPH	EQU #184C	; Wait Command (Cy = 1 Timeout ERROR)
EWAIT	EQU #184F	; Command Wait & Exit

Get Output Mode EXROM ADDRESS #1840

Retorna el Modo actual de entrada/salida en el par de registros BC
Retorna en el registro BC del Z80 el modo de salida definido para el uso de los comandos SLVM.

Este byte
TAPE/SDCARD and TS2040/PICOPT

Un valor de #0000 en el registro BC indica que la E/S corresponde al puerto de TAPE/MIC.

;-----	
; Check Local Commands	VALUE
;-----	
; SAVE "TPI:TAPE"	+0 Standart Tape Storage
; SAVE "TPI:PICOPT"	+1 PICO Virtual Printer Channel
; SAVE "TPI:SDCARD"	+2 SD Storage
; SAVE "TPI:TS2040"	+0 Standart Printer Channel
;-----	
; SAVE "TPI:FMODE=TAP"	+0 Default Mode
; SAVE "TPI:FMODE=DOS"	+4 MSDOS 3.3 (8.3 FILE NAME) / WITHOUT HEADER
; SAVE "TPI:FMODE=TPI"	+8 WINDOWS (LONG NAMES FILE NAME) / WITH HEADER
; SAVE "TPI:FMODE=FDD"	+12 CP/M - TOS (8.3 FILE NAME) / WITH HEADER
; SAVE "TPI:FMODE=RAW"	+16 WINDOWS (LONG NAMES) / WITHOUT HEADER
;-----	

Set Output Mode EXROM ADDRESS #1842

El modo de entrada/salida se debe indicar en el registro A del Z80.
; Set Output Mode (A = Output Mode)

Get TS-PICO Bios Version EXROM ADDRESS #1844

Retorna la versión de Software y Firmware en el par de registros BC
; Get Bios Version (Return Version in BC)

EXAMPLE PROGRAM IN ASSEMBLER Z80 TO OBTAIN THE MODE

ORG #9C40

```
BANKE      EQU #6499    ; Bank Enable Function
S_OFF      EQU #97C0    ; Offset to Chunk 7 Dispatcher
G_MODE     EQU #1840    ; Get Mode Routine

GMOD: CALL CALLX        ; Return Mode in BC Register
      PUSH BC           ; USE ALT Z80 Registers
      LD BC,0FF00H      ; B=DES BANK C=/HS

      PUSH AF           ; Save AF
      LD A,(VIDMOD)     ; Get Video Mode
      AND A
      JR NZ,MGBL0       ; Jump if Hi-Res Mode
      POP AF            ; Restore AF
      CALL BANKE        ; BANK ENABLE in Chunk 3.
      JR MGBL1          ; Continue

MGBL0: CALL BANKE + SECOND_OFF ; BANK ENABLE in Chunk 7.

MGBL1: POP BC           ; Return Mode in BC Register
      RET               ; Return to Caller

CALLX: EXX              ; USE ALT Z80 Registers
      LD HL,G_MODE      ; Get Output Mode Routine in EXROM
      PUSH HL
      LD HL,#FEFE       ; H=DEST BANK / L=/HS
      PUSH HL
      PUSH AF           ; Save AF
      LD A,(VIDMOD)     ; Get Video Mode
      AND A
      JR NZ,MGBL2       ; Jump if Hi-Res Mode
      POP AF            ; Restore AF
      EXX               ; Restore Z80 MAIN Registers
      CALL GOTOBK       ; GOTO BANK in Chunk 7
```

```

MGBL2:      POP AF
            EXX
            CALL GOTOBK+SECOND_OFF ; GOTO BANK in Chunk 3

```

EXAMPLE PROGRAM IN ASSEMBLER Z80 TO SEND A COMMAND BLOCK

```

00120      ORG #9C40
00130 RX_A  EQU  #1848      ; Receive Character in A  ;
00135 TPMODE EQU  #5DDB      ; TP_MODE (24027)

00140 SNDCMD: PUSH HL
00150      LD LD IX,HEADER    ; IX=Start Memory Address
00160      LD DE, 9           ; DE= Command Block LEN
00170      LD A, HTYPE        ; A= CODE TYPE
00180      ;
00190      LD (IX),A           ; Set Header Block type (SCREEN)
00200      LD A,0              ; A=Tipo de Bloque (0=HEADER FF=DATA)
00210      CALL MYSAVE
00220      LD DE, STATUS-HEADER ; DE= Block LEN
00230      INC IX
00240      LD A, BTYPE         ; A=Tipo de Bloque (0=HEADER FF=DATA)
00250      LD (IX),A
00260      INC IX ;
00270      CALL MYSAVE
00280      CALL RX_A           ; Receive Status in A
00285      LD (STATUS),A      ; Save Command Exec Status
00290      CALL WF_NPH         ; Wait for Continue

```

```

.
.
.

```

. USE SAME PROCEDURE TO SEND THE 2ND PART OF THE COMMAND

```

.
.

```

```

00300      LD A, (PORTF4)
00310      OUT (PORTF4),A      ; Restore Original HS Value
00320      LD A, (PORTFF)
00330      OUT (HREXPT),A     ; Restore Original HREXPT Value
00340      LD BC,( STATUS)
00350      LD B,0              ; Return Status Code
00360      RET

```

```

00400 MYSAVE: DI              ; Disable Interrupt
00400      PUSH AF            ; Save AF en el STACK
00410      IN A,(HREXPT)      ; A=Valor de HREXPT
00420      LD (PORTFF),A      ; Save Value for Restore Later
00430      SET 7,A             ; Set EXROM Control Bit
00440      OUT (HREXPT),A     ; Enable EXROM

```



```

00450      IN A,(DKHSPT)      ; A=Valor de DKHSPT
00460      LD (PORTF4),A      ; Save Value for Restore Later
00470      LD A,3              ; A=3 Indica EXROM en USO
00480      OUT (PORTF4),A     ; Enable EXROM in Chunk 0
00485      LD A,1              ; Set SD CARD IN USE
00490      LD (TPMODE),A     ; Flag de SD Card en USO
00500      POP AF              ; Restore AF
00510      CALL WTAPE         ; Calling W_TAPE Routine in ROM
00520      RET                ; Return

```

```

00620 PORTF4: DEFB 0 ; Save TEMP DKHSPT Port Value

```

```

00630 PORTFF: DEFB 0 ; Save TEMP HREXPT Port Value

```

```

00640 DEFB 0

```

```

00650 HEADER: DEFB 00 ; HEADER BLOCK

```

```

00660 DEFB 00 ; SAVE ID_CMND

```

```

00670 DEFB FF ; BANK NUMBER (HOME)

```

```

00680 DEFB 00 ; UNUSED

```

```

00690 DEFB 00 ; UNUSED

```

```

00700 DEFB EA ; MEMORY ADDRESS LOW

```

```

00710 DEFB 6B ; MEMORY ADDRESS HIGH

```

```

00720 DEFB 11 ; PARAMETER LEN LOW

```

```

00730 DEFB 00 ; PARAMETER LEN HIGH

```

```

00740 DEFB 43 ; CRC ENVIADO

```

```

00750 STATUS: DEFB 00 CRC COMANDO RECIBIDO (0 = OK)

```

```

00760 DEFB

```

```

EXAMPLE COMMAND SAVE "TPI:MEMBOOT" CODE 1,2

```

```

;-----
; SAVE "TPI:MEMBOOT" CODE 1,2      SET ACTIVE MEM FOR HOME & EXROM (SET PAGE 2)
;

```

```

; SAVE "TPI:MEMBOOT" CODE 1,0      SET ACTIVE MEM FOR HOME & EXROM (PAGE
DEFAULT IN CONFIG FILE)
;-----

```

```

;
; PMR1: SET ( 0 = DEFAULT / 1 = R0M / 2 = RAM / 3 = NONE)
; PMR2: PAGE #(A16-A18)
;-----

```

```

; 00      42      "B"
; 01      00      SAVE COMMAND (TADDR)
; 02      FF      BANK # (00=DOCK FF=HOME ...)
; 03      01      PMR1 SET MEMORY LOCATION (1 = R0M / 2 = RAM)
; 04      00
; 05      00      PMR2 PAGE #(A16-A18)
; 06      00
; 07      0B      BLK_LEN
; 08      00
;-----

```

```

; 09      B7      CRC
;-----

```

```

; 10      01      STATUS CODE
;-----

```

```

; 11      40      WAIT CONTINUE
;-----

```

COMMAND SECOND BLOCK

```
-----  
; 12      44      "D"  
; 13      0B      BLK_LEN  
; 14      00  
-----  
; 15      54      "T"  
; 16      50      "P"  
; 17      49      "I"  
; 18      3A      ":"  
; 19      4D      "M"  
; 20      45      "E"  
; 21      4D      "M"  
; 22      42      "B"  
; 23      4F      "O"  
; 24      4F      "O"  
; 25      54      "T"  
; 26      6B      CRC  
-----  
; 27      40      WAIT CONTINUE  
-----  
; 28      01      STATUS CODE  
-----
```

BASIC FUNCTION COMMAND IDs (ID_CMND)

```
SAVE  00  
LOAD  01  
VERIFY 02  
MERGE 03  
COPY  04  
LPRINT 05
```

NOTE:

***** It is recommended not to send messages in a different format or not defined by the protocol, as this could cause problems with system stability or produce unexpected errors.*****

CHAPTER 4

The system uses a variable in system reserved memory location whose name is TP_MODE, and is used to address the output of SLVM commands.

To select the input/output mode of the record and read functions (SAVE / LOAD / VERIFY / MERGE) using the EAR/MIC connections, you must type the following command.

The values of this variable are modified with the commands SAVE "tpi:tape" where (TP_MODE = 0) and the command SAVE "TPI:SDCARD" (TP_MODE = 1)

SAVE "TPI:TAPE"

This command sets TP_MODE = 0 and MIC and EAR ports are used for Input and Output.

To select the input/output mode of the write and read functions (SAVE / LOAD / VERIFY / MERGE) using sd memory card (Installed in the TSPICO interface) you must type the following command.

SAVE "TPI:SDCARD"

This command sets TP_MODE = 1, and the SD memory of the PICO interface is used for Input and Output.